Seed plants are vascular plants that use seeds during the dispersal stage of their life cycle. **Seeds** contain a sporophyte embryo and stored food within a protective seed coat. The seed coat and stored food allow an embryo to survive harsh conditions during long periods of dormancy (arrested state) until environmental conditions become favorable for growth. When a seed germinates, the stored food is a source of nutrients for the growing seedling. The survival of seeds largely accounts for the dominance of seed plants today.

Like a few of the seedless vascular plants, seed plants are heterosporous (have microspores and megaspores), but their innovation was to retain the spores, and not release them into the environment (see Fig. 23.18). As mentioned earlier in this chapter, the microspores become male gametophytes; these are called **pollen grains. Pollination** occurs when a pollen grain is brought into contact with the female gametophyte by wind or a pollinator. Then, sperm move toward the female gametophyte through a growing **pollen tube.** A megaspore develops into a female gametophyte within an **ovule,** which becomes a seed following fertilization.

Note that because the whole male gametophyte (a pollen grain) moves to the female gametophyte, rather than just the sperm (as in seedless plants), no external water is needed to accomplish fertilization. In essence, the less a plant relies on water for reproduction, the farther that plant can radiate onto drier land, take advantage of new resources, and become more abundant.

The two groups of seed plants alive today are gymnosperms and angiosperms. In **gymnosperms** (mostly cone-bearing seed plants), the ovules are not completely enclosed by sporophyte tissue at the time of pollination. In **angiosperms** (flowering plants), the ovules are completely enclosed within diploid sporophyte tissue (ovary), which becomes a fruit.

The first type of seed plant was a woody plant that appeared during the Devonian period (about 359.2 MYA, see Table 18.1); it has been erroneously named a seed fern. The seed ferns of the Devonian were not ferns at all; they were progymnosperms. It's possible these were the type of progymnosperm that gave rise to today's gymnosperms and angiosperms. All gymnosperms are still woody plants, but whereas the first angiosperms were woody, many today are nonwoody. Progymnosperms, including seed ferns, were part of the Carboniferous swamp forests (see the Evolution feature, "Carboniferous Forests," in Section 23.4).

Gymnosperms

The four groups of living gymnosperms (Gk. *gymnos,* "naked"; *sperma,* "seed") are conifers, cycads, ginkgoes, and gnetophytes. Because their seeds are not enclosed by fruit, gymnosperms have "naked seeds." Today, living gymnosperms are classified into more than 1,000 species; the conifers are more plentiful than the other types of gymnosperms.

Conifers

Conifers consist of about 630 species of trees, many evergreen, including pines, spruces, firs, cedars, hemlocks, redwoods, cypresses, yews, and junipers. The name *conifers* signifies plants that bear **cones,** but other gymnosperm phyla are also cone-bearing. The coastal redwood (*Sequoia sempervirens*), a conifer native to northwestern California and southwestern Oregon, is the tallest living vascular plant and may reach nearly 100 m in height. Another conifer, the bristlecone pine (*Pinus longaeva*) of the White Mountains of California, is the oldest living tree; one individual is estimated to be 4,900 years of age.

Vast areas of northern temperate regions are covered in evergreen coniferous forests (Fig. 23.17). The tough, needlelike leaves of pines conserve water, because they have a thick cuticle and



a. Pine


b. Spruce


c. Cypress

Figure 23.17 Conifers. **a.** Pine trees are the most common of the conifers. The pollen cones (male) are smaller than the seed cones (female) and produce pollen. Other conifers include (**b**) the spruces, which make beautiful Christmas trees, and (**c**) the cypress, which can be found in northern temperate regions and is a popular landscape tree. (a): (seed cones): Steven P. Lynch; (pollen cones): Ricochet Creative Productions; (b): Ed Reschke/Stone/Getty Images; (c): Steffen Hauser/botanikfoto/Alamy Stock Photo

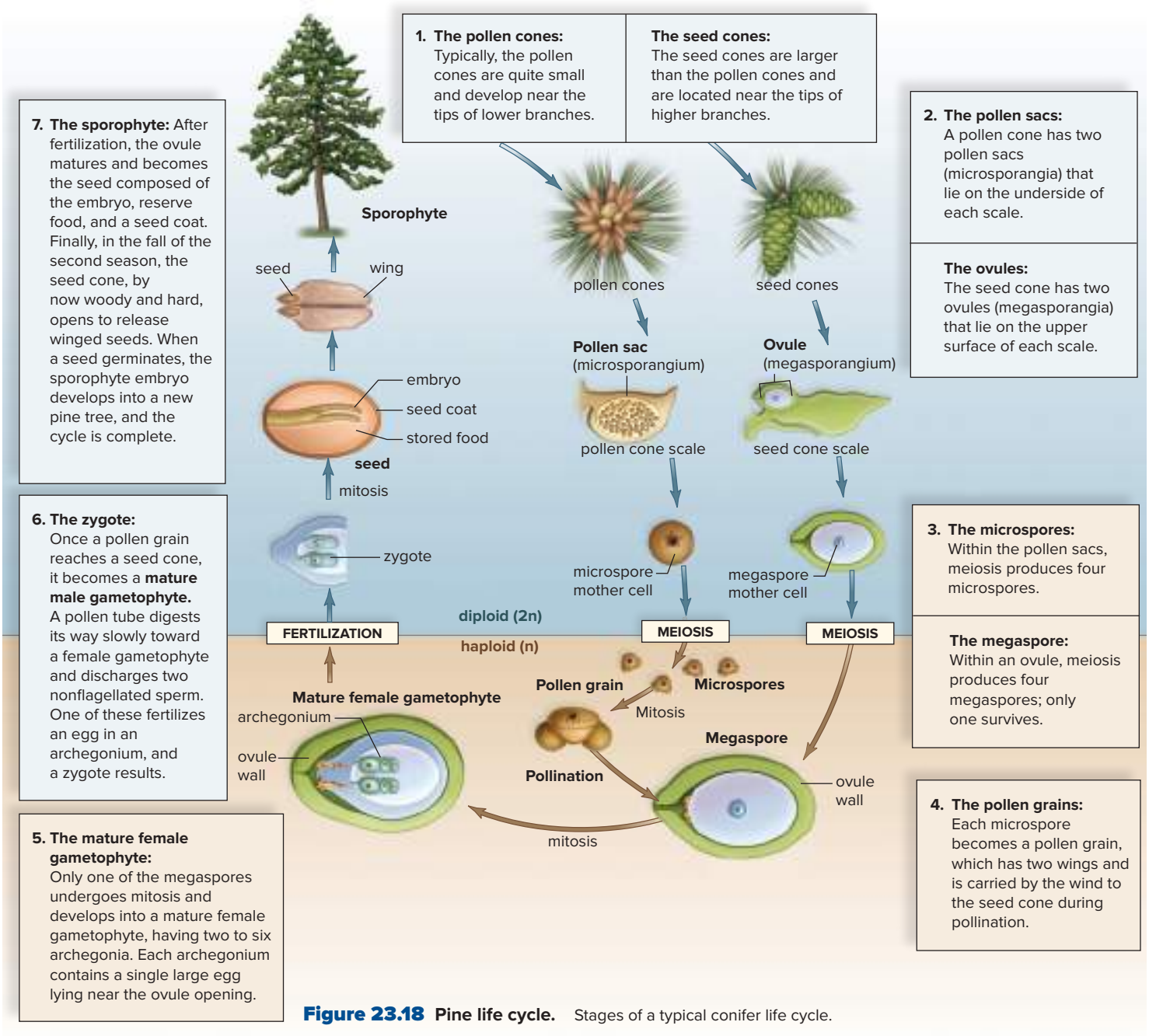


Figure 23.18 Pine life cycle. Stages of a typical conifer life cycle.

recessed stomata. Note that in the life cycle of the pine (Fig. 23.18) the sporophyte is dominant, pollen grains are windblown, and the seed is what is dispersed. Conifers are **monoecious,** which means that a single plant carries both male and female reproductive structures.

Pines, in particular, are well-known for their beauty and pleasant smell; they make attractive additions to parks and gardens. Pines have medicinal value in that the needles and bark are rich in vitamins A and C and can be used to create teas that ease symptoms of a cold or cough. Large pine seeds, called pine nuts, are sometimes harvested for use in cooking, and pine oil is used to scent a number of household and personal care products, such as room sprays and deodorant.

Cycads

Cycads include ten genera and 300 species of distinctive gymnosperms. The cycads are native to tropical and subtropical forests. *Zamia pumila,* found in Florida, is the only species of cycad native to North America. Cycads are commonly used in landscaping. One species, *Cycas revoluta,* referred to as the sago palm, is a common landscaping plant. Their large, finely divided leaves grow in clusters at the top of the stem, and therefore they resemble palms or ferns, depending on their height. The trunk of a cycad is unbranched, even if it reaches a height of 15–18 m, as is possible in some species.

Unlike conifers, cycads have pollen and seed cones on separate plants. The cones, which grow at the top of the stem surrounded by the leaves, can be huge—more than a meter long with a weight of

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